

1. Difference between Fertility and Fecundity

Aspect	Fertility	Fecundity
Definition	Number of live births women have (actual reproduction)	Physiological capacity to produce a live birth (potential reproduction)
Nature	Measures actual reproductive performance	Measures biological capability
Measured By	Number of live-born children	Ability to conceive and carry pregnancy to term
Dependency	Influenced by social, cultural, economic, environmental, and health factors	Depends on biological and physiological health
Example	A woman has 3 children → fertility = 3	A healthy woman capable of bearing children is fecund
Limitation	A fecund woman may not be fertile if she uses contraception, abstains, or delays childbirth	Infecundity (sterility) means no biological capacity to bear children

2. Crude Birth Rate (CBR) Formula & Example

$$\text{CBR} = \frac{\text{Number of live births in a year}}{\text{Mid-year total population}} \times 1000$$

Example:

If there are 21,500 births in a population of 1,000,000:

$$\begin{aligned}\text{CBR} &= \frac{21500}{1000000} \times 1000 \\ &= 21.5 \text{ per 1,000 population}\end{aligned}$$

3. General Fertility Rate (GFR) Formula & Example

$$\text{GFR} = \frac{\text{Number of live births in a year}}{\text{Number of women aged 15–49}} \times 1000$$

Example:

If there are 91,000 births among 1,000,000 women aged 15–49:

$$\begin{aligned}\text{GFR} &= \frac{91000}{1000000} \times 1000 \\ &= 91 \text{ per 1,000 women aged 15–49}\end{aligned}$$

4. Age-Specific Fertility Rate (ASFR) Formula & Example

$$\text{ASFR} = \frac{\text{Number of births to women in age group}}{\text{Number of women in that age group}} \times 1000$$

Example (Age 20–24 in USA, 2008):

Births = 1,052,184

Women = 10,215,379

$$\text{ASFR} = \frac{1052184}{10215379} \times 1000$$

= 103.0 births per 1,000 women

5. Total Fertility Rate (TFR) Formula & Example

$$\text{TFR} = \sum \text{ASFRs} \times \text{Width of age interval}$$

Example (USA, 2008):

Sum of ASFRs = 0.417 (rates in proportion, not per 1,000)

Width of interval = 5 years

$$\text{TFR} = 0.417 \times 5 = 2.09 \text{ births per woman}$$

6. Theories of Fertility (9 Theories)

a. Demographic Transition Theory

- **Proposed by:** Warren S. Thompson (1929) and Frank W. Notestein (1945).
- **Concept:** Describes the transformation of a population from high birth and death rates to low birth and death rates as a country moves from a pre-industrial to an industrialized economy.
- **Four classic stages:**
 1. **Pre-transition:** High birth rate & high death rate → little or no population growth. (No modern example.)
 2. **Early transition:** Birth rate remains high, death rate falls due to better healthcare and sanitation → rapid population growth. (Many developing countries.)
 3. **Late transition:** Death rate remains low, birth rate declines → slower growth. (E.g., China, Indonesia.)
 4. **Post-transition:** Both birth and death rates are low → population growth stagnates or declines. (E.g., most European countries.)
- **Implication:** As countries develop, fertility falls naturally after mortality declines.

b. Wealth Flows Theory

- **Proposed by:** John C. Caldwell (1976).
- **Concept:** Fertility depends on the direction of intergenerational flows of wealth.
 - **If wealth flows from children to parents:** Children contribute economically (labor, income, care) → families prefer large numbers of children.
 - **If wealth flows from parents to children:** Children are financially dependent → families prefer fewer children.

- **Modern implication:** In industrial societies, raising children is costly, so fertility is lower.

c. Proximate Determinants Theory

- **Introduced by:** Kingsley Davis & Judith Blake (1956); refined by John Bongaarts (1982).
- **Concept:** Fertility is influenced by intermediate variables that directly determine the number of births.
- **Bongaarts' 7 determinants:**
 1. Marriage and marital disruption.
 2. Contraceptive use and effectiveness.
 3. Induced abortion.
 4. Duration of postpartum infecundability (e.g., breastfeeding).
 5. Waiting time to conception.
 6. Risk of intrauterine mortality.
 7. Onset of permanent sterility.
- **Implication:** Policies targeting these determinants can influence fertility directly.

d. Human Ecological Theory

- **Concept:** Fertility patterns are shaped by the interaction between human populations and their environment.
- Population density, resource availability, and ecological constraints influence reproductive decisions.
- For example, in resource-scarce areas, fertility might be limited by environmental carrying capacity.

e. Political Economic Theory

- **Concept:** Fertility is influenced by political systems, economic structures, and policies.
- Factors such as labor market conditions, welfare policies, women's employment opportunities, and taxation can affect decisions on childbearing.
- Example: Countries with generous parental leave may have higher fertility compared to those without such benefits.

f. Feminist Theory

- **Concept:** Emphasizes gender relations and women's empowerment in shaping fertility decisions.
- Access to education, reproductive rights, and autonomy allow women to control family size.

- Fertility decline is linked to women's participation in the workforce and decision-making within families.

g. Bio-Social Theory

- **Concept:** Fertility results from an interaction between biological capacity and social norms.
- Even if a woman is biologically capable of having children (fecund), social, cultural, or personal factors may limit the number of births.
- Example: Cultural expectations about ideal family size influence actual fertility behaviour.

h. Relative Income Theory

- **Proposed by:** Richard Easterlin (1970s).
- **Concept:** Fertility is determined by the ratio of actual income to expected income.
 - If actual income meets or exceeds expectations → higher fertility.
 - If actual income falls short of expectations → lower fertility.
- Economic uncertainty can lead to postponement or reduction of childbearing.

i. Diffusion Theory

- **Concept:** Fertility decline occurs as new ideas, technologies, and behaviors spread across societies and social groups.
- Examples include adoption of contraception, changing attitudes toward family size, and influence of mass media.
- Social networks, urbanization, and migration accelerate the spread of low-fertility norms.